

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination 2025-26 (NEP)

PHYSICS**Paper Code: PHYMJ41 / PHYMN41 : Electromagnetic Theory**

Ref. No.: Conf/Even/N/2025-26/58

Time: $2\frac{1}{2}$ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. All symbols have their usual meanings unless specified otherwise.

Question 1. Answer any four of the following questions:

[4 × 5 = 20]

- Define the gradient of a scalar function. Find the gradient of the function $f(x, y, z) = x^2y^3z^4$.
- What is the magnetic field outside an infinite solenoid with current carrying wires with current density I wrapping it?
- For the vector $\vec{v} = (2xz + 3y^2)\hat{a}_y + (4yz^2)\hat{a}_z$, calculate the line integral for the closed square path in $y - z$ plane.
- What are Lorentz and Coulomb gauges?
- Find the skin depth of copper whose conductivity $\sigma = 5.8 \times 10^7$ mho/m, $\mu = \mu_0$ at frequency 10^8 Hz.
- Find the electric field at a distance d from an infinite sheet with uniform surface charge density σ .

Question 2.

- What are the Lorentz transformations? Show that the combination $x^2 - c^2t^2$ does not change under Lorentz transformations. [7½]
- Show that wave equation with velocity c , does not change under Lorentz transformations. [5]

Question 3.

- Consider a reference frame R_2 moving in a velocity v_2 with respect to a reference frame R_1 , which in turn moves in a velocity v_1 with respect to another reference frame R_0 . What is the velocity of R_2 with respect to R_0 ? [5]
- A stick of length 34 cm (in rest frame) is moving at the speed $v = \frac{15}{17}c$ with respect to observer A. What will be the length of the stick according to A? [5]
- A graviton (the particle carrying gravitational force) has vanishing rest mass. What would be its velocity with respect to ground? [2½]

Question 4.

- (a) Write down the Maxwell's equations in vacuum. Which of these equations satisfy vector identities? [4]
- (b) Show that the electric field obeys a wave equation. [4½]
- (c) Write down Maxwell's equations in a medium. How do they simplify for a linear media? (derivation not required) [4]

Question 5. Derive the expressions for the reflection and transmission coefficients when an electromagnetic wave incident obliquely on a boundary separated by two lossless dielectrics for the case when electromagnetic wave electric field is perpendicular to the plane of incidence. [12½]

Question 6.

- (a) Discuss the propagation of electromagnetic wave in conducting media and derive attenuation and phase constants. [9]
- (b) Calculate the intrinsic impedance of free space. [3½]

Question 7. State the Stoke's theorem for a vector field. Suppose vector field $\vec{v} = xy\hat{a}_x + 2yz\hat{a}_y + 3zx\hat{a}_z$. Check the Stoke's theorem using a triangular surface with vertices $(0, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$ in $y - z$ plane. [12½]

Parameters: $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$, $e = 1.6 \times 10^{-19} \text{ C}$.
